

ASSIGNMENT 4.3

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

No.	BIOS / POST Error Message	Typical Hardware / Configuration issue
1.	CMOS checksum error defaults loaded	CMOS battery weak/dead or BIOS settings corrupted; BIOS reset to defaults.
2.	BIOS ROM checksum error system halted	Corrupt BIOS firmware or faulty BIOS chip; sometimes after bad flash or power loss.
3.	Keyboard error or no keyboard present	Keyboard not detected: loose cable/USB, faulty keyboard, or wrong BIOS keyboard setting.
4.	CPU fan Error / CPU fan not detected	CPU fan not spinning, disconnected or connected to wrong header; BIOS thinks CPU cooling is missing.
5.	No boot device found/boot device not found	No bootable drive detected: loose SATA/power cable, failed SSD/HDD, wrong boot order, or missing OS.

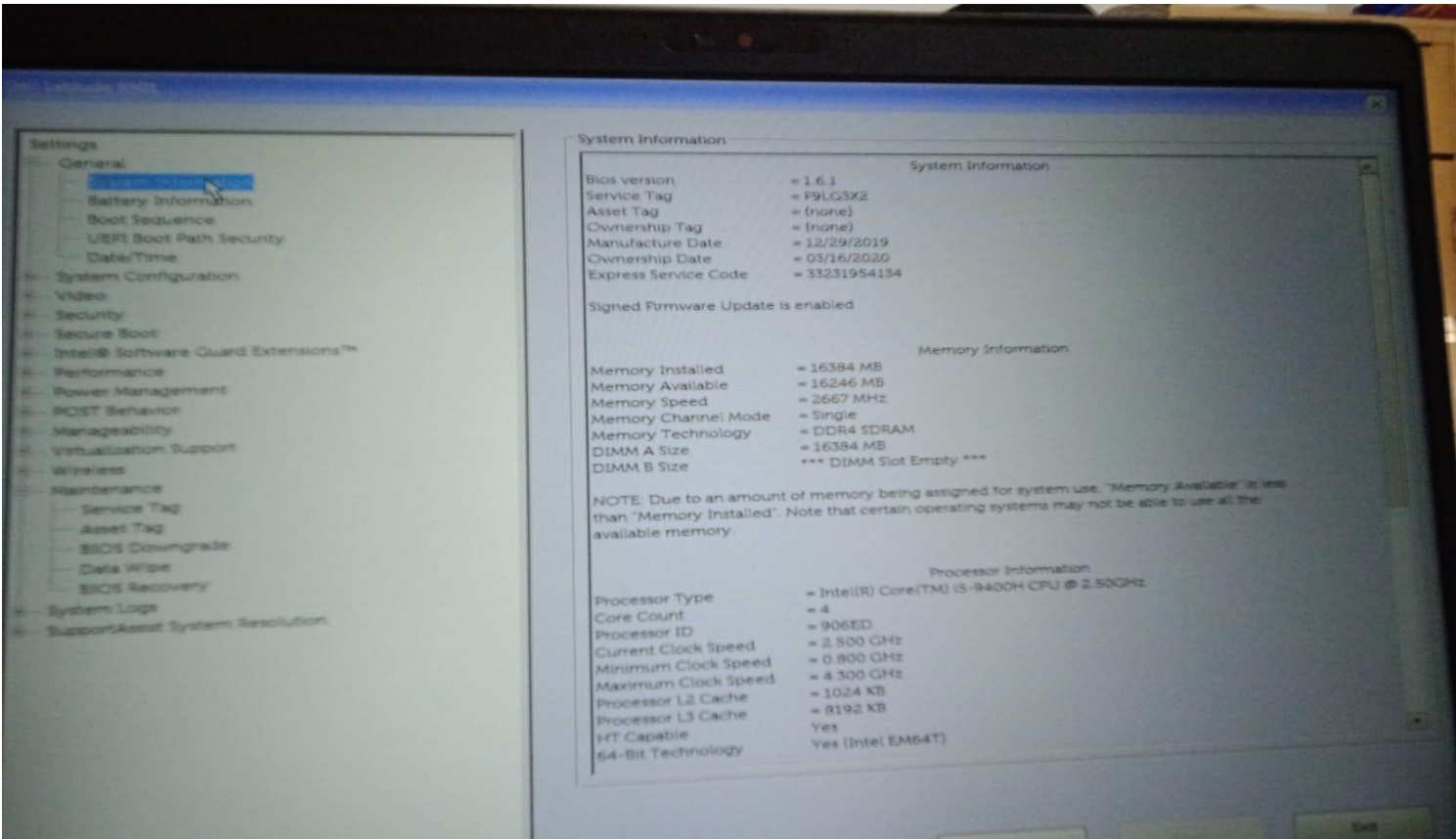
2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Meaning (AMI BIOS): for AMI BIOS, short beeps typically mean: “base memory read/write test error” – a RAM problem in the first 64KB of system memory.

Two troubleshooting steps:

- **Reset/Test RAM:** power off and unplug PC. Open case, remove RAM sticks, clean contacts gently, and reinstall firmly. If multiple sticks, try one stick at a time in different slots to find a bad module/slot.
- **Clear CMOS / Reset BIOS:** power off, unplug, and either remove CMOS battery for 30 seconds then reinsert, or use the CLR_CMOS jumper as per motherboard manual. Boot again; if it POSTs, reconfigure BIOS settings.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



**4.You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset. **

What it Means:

This message means the BIOS settings stored in CMOS memory do not match the checksum the BIOS expects. Most common cause: CMOS battery (CR2032 coin cell) is weak or dead, so settings are lost when power is off. As a result, BIOS loads default settings so the system can still boot.

How to Resolve (Real System):

- Replace CMOS battery: power off and unplug PC. Open case, locate silver coin-cell battery (usually CR2032) on motherboard. Remove old battery, wait 30 seconds, then insert new CR2032 with + side up.
- Reset / reconfigure BIOS settings: power on and press Del / F2 (as per motherboard) to enter BIOS if needed to set correct date/time, proper boot order, and any custom settings (XMP, fan curves, etc.).

- Verify Fix: restart PC a few times and ensure the “CMOS checksum error” message no longer appears. If error return soon after, re-check battery seating or try another new CR2032.